The opening prompt and the five-volume workflow

COPY INTO EMERGENT FIRST

Build Tejas Exam Studio, a polished blue-themed exam practice platform for Tejas and a large student community. The name is provisional and must be configurable.

Copy-ready master instruction

Act as a senior product engineer, exam-domain analyst, accessible interface designer, and quality owner. Build a working application in reviewable increments. Students must take JEE Main, BITSAT, WBJEE, and COMEDK practice examinations using approved, year-specific exam profiles. Label WBJEE as a digital practice adaptation of an official OMR examination. Administrators will upload question PDFs and separate answer-key or solution PDFs. Import these into an editable review workflow, publish only approved papers, preserve question-level provenance, and present correct, reproducible results with at least 40 distinct analyses. Volume 4 specifies 60 analyses and the evidence each requires.

Use a simple, beautiful, restrained blue interface. Prioritize correct question rendering, answer persistence, scoring, timer behavior, recoverable failures, and useful review over decorative effects. Design for a large registered audience, and prove simultaneous capacity through measured tests rather than claims. This is an independent practice service: never imply official affiliation or that platform percentile is an official exam percentile.

Supply the volumes in this order

First load this product and UX volume to establish the application purpose and screens. Next apply Volume 2 for official evidence, exam profiles, state transitions, and scoring rules. Then apply Volume 3 for PDF ingestion, question data, solutions, tagging, and editorial quality. Apply Volume 4 for measurement definitions, results, cohort statistics, and recommendations. Finally apply Volume 5 for architecture, security, operations, implementation stages, and release verification.

Do not paste all five PDFs into one uncontrolled generation. In each stage ask Emergent to summarize the binding requirements, identify unresolved decisions, implement the agreed slice, show actual evidence, and update a requirement traceability checklist and changelog. Ask it to read the relevant pages again before changes that affect that area. Clear purpose, core features, and design guidance follow Emergent's prompting recommendations [V1-S3].

Resolve conflicts explicitly

Official source evidence outranks examples in this pack for official exam behavior. An approved, versioned exam profile implements that evidence. Content publication and analytics must follow that profile. If source evidence is unclear, show a review task and keep that profile unpublished. Product examples never override a verified rule. Development shortcuts must not silently become production behavior.

ACCEPTANCE / PRECAUTIONS

- Demonstrate one complete student attempt and one complete administrator publishing flow before adding breadth.
- Maintain a decision log with source, owner, version, date, and affected acceptance tests.

Product promise, boundaries, and honest claims

PRODUCT CONTRACT

The product succeeds when a student can complete a dependable practice attempt and understand what to improve next without needing help to interpret the interface.

The promise to the student

Offer faithful practice for supported exam profiles, clear custom practice for selected chapters, readable questions and solutions, and analysis that identifies concrete actions. Preserve the distinction between an exam simulation and a teaching session. A simulation follows its approved navigation and submission rules; a learning session may reveal hints or explanations only if explicitly enabled. Every attempt shows its mode before starting and in its final report.

Treat Tejas as an initial student persona, not as a special privileged account. His restricted chapter selection and integer-only practice requirements are supported through general configuration that any permitted student can use. Keep personal student names out of public marketing copy and shared examples unless they are intentional, approved content.

Boundaries that prevent misleading behavior

Do not promise official exam scores, admission chances, guaranteed ranks, or psychological diagnoses. Do not call a self-selected practice cohort the national population. Never produce a percentile when the metric definition lacks a suitable cohort, eligibility rules, or minimum evidence. Provide an unavailable state with a practical explanation. Predicted outcomes, when supported later, must have a model version, uncertainty, validation, and a clearly visible prediction label.

Question quality and answer correctness are editorial responsibilities. AI may suggest extraction, tags, or explanations, but an attractive interface does not validate those outputs. Show content quality status to administrators and prevent unapproved content from appearing in student attempts.

Scope for the first working release

Deliver approved exam profiles, PDF import review, a reliable exam shell, scoring, question-wise review, basic progress history, and a meaningful subset of the Volume 4 analytics. Build the remaining analytics as available data permits. The required launch threshold is at least 40 correctly implemented, distinct analyses with explicit unavailable states; 60 is the complete specification. Repeated visualizations of the same number do not count as separate analyses.

Acceptance language for Emergent

Copy this requirement: Every visible product claim must correspond to a tested capability or a clearly stated limitation. Use actual application records for screenshots and demos. Label demonstration data. Produce a short release note that states what works, which exam profile versions were verified, and which planned capabilities remain unavailable.

ACCEPTANCE / PRECAUTIONS

- The landing page says independent practice platform and names supported profiles accurately.
- Results distinguish measured values, derived values, unavailable analyses, and predictions.

Modes, personas, and role boundaries

ONE PRODUCT, EXPLICIT MODES

Use a small set of understandable modes and roles. Avoid granting privileges because a user reaches an administrator-looking URL.

Student modes

Full Simulation loads an approved complete paper, its exam profile, its time budget, and its allowed interaction rules. For WBJEE, this is a digital practice adaptation; never market it as an official CBT replica. Custom Practice lets a student select approved subjects, chapters, question types, count, and a supported time setting. Label integer-only JEE-style practice as Custom Practice when its composition differs from the official profile. Topic Learning is an optional later mode with hints and immediate explanations. Its attempts must be excluded from simulation comparison cohorts unless a metric specifically includes them.

Each attempt stores mode, paper revision, exam-profile version, content release, language, accommodation setting, and cohort eligibility. These identifiers stay fixed after start. A saved attempt cannot acquire different scoring rules because an administrator updates the profile while it is in progress.

Personas and the tasks they need

A first-time student needs a quick orientation, one suitable paper, and confidence that answers save. A returning student needs unfinished attempts, previous errors, and a small next-action list. A parent or mentor may need an explicitly shared progress view, but sharing is optional and permission-controlled. A content editor needs PDF comparison, unresolved import issues, question editing, and validation. An exam-rule reviewer needs evidence and profile approval. An operator needs incident visibility and support tools with limited access.

Role design

Define Student, Content Editor, Content Reviewer, Exam Profile Reviewer, Support Operator, and Platform Administrator as permission sets. Small teams may assign multiple sets to a person, but each privileged action still requires its relevant permission. Separate uploading from publication approval by default. Support personnel cannot browse complete student answer histories merely because they can resolve a login ticket.

Invite and remove administrator access through an auditable interface. Require additional authentication for sensitive actions as specified in Volume 5. Never expose role changes, answer keys, or unpublished solutions to public clients.

Mode selection interaction

Present a simple choice with a one-sentence explanation and clear consequence. A simulation card states the profile and separate result and solution release conditions, including any scheduled embargo. A custom-practice card says that chapter and composition settings differ from a full exam. Show a confirmation if changing a configured practice mode would discard selections.

ACCEPTANCE / PRECAUTIONS

- A student token cannot publish a paper, read unreleased solutions, or assign roles.
- Reports and comparison filters always show or retain the attempt mode.

The restrained blue design system

EXACT VISUAL TOKENS

Build a consistent interface from tokens. The exam workspace should feel calm, legible, and predictable for a long attempt.

Color tokens and their purpose

Use primary #1D4ED8, primary-hover #1E40AF, primary-soft #EFF6FF, page-background #F5F8FD, surface #FFFFFF, text-strong #0F172A, text-muted #475569, border #CBD5E1, success #166534, success-soft #F0FDF4, warning #92400E, warning-soft #FFFBEB, danger #B91C1C, and danger-soft #FEF2F2. Use focus-ring #2563EB with a visible outer offset. Verify every actual text/background pairing and state rather than assuming a token alone passes accessibility.

Status must combine color with a label, symbol, or shape. Marked-for-review, answered, and unanswered states must remain distinguishable in grayscale. Avoid gradients behind mathematical content, low-contrast pastel text, pulsing urgency effects, and decorative backgrounds in the exam area.

Typography and spacing

Use Inter when available, with system-ui and sans-serif fallbacks. Use a separate tested math renderer and its compatible fonts. Desktop body text is 16 px with 1.5 line height; question content defaults to 18 px with 1.55 line height and offers a readable size setting. Small metadata is 14 px. Headings use 24, 30, or 36 px only where hierarchy warrants it. Avoid condensing body text to fit a card.

Use spacing steps 4, 8, 12, 16, 24, 32, 48, and 64 px. Cards have 12 px corner radius; controls have 8 px radius; dialogs have 16 px radius. Default card padding is 24 px on desktop and 16 px on narrow screens. Use light borders and restrained shadows. The question canvas is the visual center.

Layout tokens

Set reading content to a comfortable maximum width of 720 px, general dashboard content to 1200 px, and the desktop exam palette to roughly 280 px where space permits. Treat these as layout targets that must adapt without clipping. Use a consistent header, breadcrumb, page title, and action placement. Avoid displaying more than one competing primary action in the same decision area.

Implementation direction

Define the tokens centrally as CSS variables and application theme values. Components consume tokens rather than hard-coded ad hoc colors. Ship light mode first; dark mode is optional only after question figures, math, status colors, contrast, and screenshots have all been verified.

ACCEPTANCE / PRECAUTIONS

- Check the complete theme against WCAG 2.2 contrast and focus expectations [V1-S1].
- At 200% zoom, controls remain visible and no question or answer is clipped.

Components and predictable interaction

UI CONTRACT

A component is complete only when its loading, focus, disabled, error, and success states are understandable.

The shared component inventory

Create buttons, text fields, selects, numeric answer fields, radio options, checkboxes, tabs, dialogs, tooltips, notices, skeletons, badges, empty states, tables, pagination, paper cards, question cards, palette cells, save indicators, and metric cards. Document permitted variants and accessibility semantics. Keep student and administrator interfaces visually related while allowing denser editorial tools.

Buttons must have an action label: Start simulation, Continue attempt, Save and next, Review paper, or Publish approved paper. Avoid generic Submit wherever it hides what will happen. Show a loading label only when a request is actually running. Prevent duplicate actions through the mutation contract, with interface feedback that does not depend solely on disabling a button.

Interaction details

A modal dialog announces its purpose, receives focus, traps focus while open, and returns focus to its trigger when dismissed. Keep destructive actions visually distinct and require an intentional choice for the affected record. Menus close predictably with Escape. Tooltip content is supplemental; instructions required to complete an action must be visible without hovering.

For editable fields, associate labels, guidance, and validation errors programmatically. Preserve entered values after a failed request. Set focus to the first relevant error when a form cannot proceed, without destroying the rest of the input. Keep inline feedback close to the control; use a summary when there are several failures.

Exam-specific exceptions

Use stable option positions throughout an attempt. Do not animate selected options, shuffle answers after a revisit, or show correctness before the mode permits it. A numeric field must reflect the allowed answer syntax of the approved profile; it must not infer permissibility from a browser keyboard alone. Keyboard shortcuts are discoverable and configurable, and must not intercept text entry.

Copy-ready implementation requirement

Build a component gallery route accessible only to the team. Show every interaction state with actual semantics, not screenshots. Include a long mathematical stem, a tall figure, mixed-language content, a large option, a failed save, and a narrow viewport. Reuse these fixtures in device checks. Document any deliberate differences between the official simulation controls and ordinary dashboard controls.

ACCEPTANCE / PRECAUTIONS

- Keyboard users can activate, exit, and revisit every interactive component.
- Disabled controls explain their condition when the user needs that explanation; success messages reflect server-confirmed outcomes.

Routes, navigation, and permission guards

SITEMAP THAT IMPLEMENTS THE PRODUCT

Create clear destinations and preserve context through navigation. A route list is not sufficient without data access and state behavior.

Student routes

Use / for the public landing page; /sign-in and /onboarding for access and orientation; /dashboard for the student home; /exams for the catalog; /exams/:paperId for details; /practice/new for custom configuration; /attempts/:attemptId/instructions for readiness; /attempts/:attemptId for the exam shell; /attempts/:attemptId/results for the overview; and /attempts/:attemptId/review/:questionId for question-wise review.

Use /analytics for longitudinal views, /library for bookmarked and previously missed questions, /planner for next actions, /profile for preferences and permitted data controls, and /help for product support. Keep URLs stable enough that a student can reopen a permitted result link. A shared link must apply its own authorization policy and never inherit accidental access from a guessed identifier.

Administrator routes

Provide /admin/papers, /admin/imports/:importId, /admin/questions/:questionId, /admin/profiles, /admin/reviews, /admin/cohorts, /admin/support, and /admin/audit with role-specific guards. Route guards improve navigation but server authorization controls every record and action. Editors do not gain access to other organizations or all student records by changing a URL.

Navigation behavior

The student sidebar groups Home, Exams, Results, Question Library, and Planner. Hide optional unfinished areas. Keep the active destination visible and retain catalog filters when returning from exam details. The exam shell uses its own focused layout; ordinary marketing navigation must not distract from an active attempt. Navigating away invokes only the warnings and continuation behavior permitted by that profile.

Handle not-found, unavailable, forbidden, and signed-out states separately. If an expired session can be renewed, preserve the intended safe destination. Do not put sensitive answer values, private student identifiers, or access tokens into URLs.

Deep-link acceptance

Copy this direction: Test each protected route as an owner, another student, a signed-out visitor, and each relevant administrator role. Verify the API denies forbidden data before it reaches the client. Opening an already-submitted attempt redirects to its permitted report; opening an unavailable paper gives an explanatory page rather than a broken exam shell.

ACCEPTANCE / PRECAUTIONS

- Browser Back restores useful catalog or review context without restarting an attempt.
- A forged URL cannot reveal another student report or unpublished answer key.

Onboarding, sign-in, and a useful first session

REDUCE FRICTION WITHOUT HIDING CHOICES

A new student should understand the platform and reach a suitable paper quickly. Collect only information with a clear product purpose.

The onboarding sequence

Start with account access, then ask for a display name, target exams, preferred interface language, and an optional preparation stage. Explain why each field is useful. Keep examination year selectable rather than inferring it from age or school year. Allow students to skip optional personalization and change it later. Do not request marks, school, phone number, or demographic details simply to make onboarding look complete.

Offer two first actions: Try the short interface rehearsal or Browse papers. The rehearsal uses demonstration questions, records no competitive percentile, and clearly says it is not an official mock. Returning students see Continue attempt only when a continuable attempt exists under its profile rules.

Account behavior

Use the authentication mechanisms implemented and verified in Volume 5. Provide clear request, confirmation, expired-link, retry, and signed-out states. Avoid revealing whether a particular email belongs to a student through uncontrolled account lookup. Keep sign-in feedback consistent and specific enough to help the person using it. Support deliberate sign-out and explain what happens to a running attempt before the student takes that action.

Treat a student account as personal. If mentor views are included later, the student initiates a scoped sharing invitation, sees what will be shared, and can revoke it. No public profile or leaderboard participation is assumed at registration.

Preferences and responsible defaults

Store readable question size, language, motion preference, notification choices, and permitted accessibility preferences. Provide an explicit first-session privacy notice with links to the applicable policy. Handling minors, consent, and data retention requires the applicable implementation review described in Volume 5; do not invent an age threshold or call a checkbox legally sufficient.

Default to a quiet dashboard with no autoplay, no notification barrage, and no pressure countdown unrelated to an exam. Do not turn the first login into a mandatory purchase or a multi-screen survey.

First-session quality

Present the chosen exam, a suggested available paper, its mode, and its expected time commitment. If suitable approved content is unavailable, say so and show alternatives without silently switching examination formats. Show a help link near the first start action.

ACCEPTANCE / PRECAUTIONS

- A new account can start a suitable available paper without completing optional fields.
- Refreshing onboarding, signing out, and returning do not create duplicate accounts or discard valid preferences.

Catalog, search, and paper discovery

HELP STUDENTS PICK THE RIGHT ATTEMPT

Make the exam catalog informative enough to choose a paper without exposing unreleased questions or solutions.

Catalog structure

Provide tabs or filters for exam family, profile year, Full Simulation or Custom Practice, subject, chapter coverage, duration, language, difficulty descriptor, and availability. Display only meaningful filters backed by indexed approved data. Use an obvious Clear filters action and preserve selections in safe URL parameters. Search matches paper titles and approved topic labels; it does not leak private solution text.

Each paper card displays title, examination family, profile version or year, mode, subject coverage, question count, permitted duration summary, content quality status, and the action available to this student. A published card can say Reviewed. Avoid a decorative Verified badge unless the exact verification scope is explained.

Selecting and comparing papers

A paper detail link is separate from Start. Show completed-attempt count and a clearly labeled latest personal score where useful. A student who already started the same assigned attempt sees Continue, not another ambiguous Start button. An optional Compare view may show up to three papers using comparable attributes; it must not imply their raw marks are directly comparable across examination families.

Display difficulty labels as editorial or empirical according to their source. Empirical labels need their cohort and version; new papers can have an editorial label or show Not yet calibrated. Do not fabricate the number of students who took a paper.

Empty and unavailable states

When filters match nothing, retain the filters, explain the mismatch, and offer to remove the most restrictive selection. When the catalog is empty, say No published papers are available yet. If access is required, explain the applicable entitlement accurately. If an import is still under review, hide it from students or label it as unavailable without offering Start.

Search loading must not resemble an empty result. A failed request shows Retry while preserving the query. Keep prior results visibly stale only if that state is clear and safe.

Implementation direction

Copy this requirement: Build catalog search with stable pagination or a tested incremental-loading strategy, keyboard-operable filters, and understandable result counts. Test a large paper list with long names and many chapters. Ensure the backend applies publication and access filters before returning records, rather than filtering restricted records only in the browser.

ACCEPTANCE / PRECAUTIONS

- A student can identify exam family, mode, and time commitment before starting.
- No unpublished questions or solutions are included in catalog responses.

Exam details and pre-start readiness

MAKE THE COMMITMENT UNDERSTANDABLE

The exam detail page should answer what the student will take, under which rules, and what is needed for a reliable attempt.

Information before the start action

Show the paper title, independent practice disclaimer, exam family, year-specific profile, mode, language, subject and section structure, question count, duration summary, scoring summary, and publication revision. All numerical rules come from Volume 2 and the approved configuration. Link to the expanded instructions. Custom-practice details show selected chapters and deviations from a full simulation prominently.

Explain whether the student can pause, resume after disconnection, move across sections, revisit answers, and change answers after advancing. Use the actual approved profile behavior. Avoid universal phrases such as Resume anytime when only some modes permit it. State when solutions become available and which events end the attempt.

Readiness check

Check whether the browser can load the question assets and communicate with the attempt service. Offer a short demonstration of selecting an option, entering a numeric answer, marking review, and moving between permitted questions. Show recommended device conditions as guidance; do not require a desktop if the product supports phones. Keep the readiness check separate from a guarantee that the connection will never fail.

Show current connectivity and a tested save-status example. If prerequisites fail, describe the failing requirement and provide a retry path. Do not allow a failed readiness check to consume an examination timer unintentionally. Accommodations must be selected and authorized before the attempt snapshots its profile.

Starting without duplicate attempts

The start button creates or resumes the appropriate attempt through an idempotent server action. Show Creating attempt and move to instructions only after confirmation. A double click, refresh, or repeated network request must not create duplicate assigned attempts. If a practice policy intentionally allows another attempt, label it Start another attempt and describe any comparison consequences.

Version and fairness disclosure

Store the displayed paper revision and rule version with the attempt. If the paper becomes unavailable before start, show a clear explanation and return to discovery. If a published correction affects completed attempts, use the revision and regrade procedure in Volumes 3 and 5; do not quietly replace the report that the student previously saw.

ACCEPTANCE / PRECAUTIONS

- The shown rules match the configuration that the server snapshots at start.
- A failed start request preserves selections and offers a safe retry without duplicate records.

Instructions and the interface rehearsal

CONFIRM UNDERSTANDING, NOT JUST A CHECKBOX

Use concise instructions with expandable detail. Important consequences must remain visible before the timer begins.

Instruction page composition

Begin with the mode, examination profile, and duration. Present answer types and scoring, navigation restrictions, save behavior, review markers, timing, submission, and permitted continuation as separate readable blocks. Include an accessible legend for palette statuses. Numerical and transition details are supplied by Volume 2. Show a short section summary even when the full rule text is available below.

Place the primary Begin timed attempt action at the bottom of the required instruction content, with the time consequence beside it. Do not preselect a rule-acceptance control. Agreement records acknowledgment of these instructions; it is not a substitute for an actual working save and timer system. If rules have materially changed since a previous attempt, highlight the change.

Rehearsal behavior

Offer a separate untimed rehearsal using obvious demonstration content. Practice choosing an option, clearing a permitted answer, entering a numeric value, marking review, opening the palette, and confirming a demonstration submission. The rehearsal must reflect the interaction affordances of the selected profile, while avoiding false claims that it reproduces every official examination-room condition.

After rehearsal, return to instructions without carrying answers into the real attempt. Explain any control that differs by mode, such as hints being available in learning practice but not simulation. The student may skip rehearsal after understanding the controls.

Accessibility and clarity

Provide a textual status legend, visible focus, readable tables, and an accessible order for screen readers. Do not express review status using colors alone. Keep rule sentences short. Explain ambiguous terms such as attempted, submitted, and marked for review using the approved state definitions. Provide a translated interface where approved, without automatically translating mathematical question content.

Instruction acceptance evidence

Copy this acceptance request into Emergent: Demonstrate instructions for each supported examination family and for custom integer practice. Show that starting begins the correct timer only after the server transition succeeds. Test back navigation, refresh, delayed network response, double activation, and session expiry. No request failure may leave the student believing the exam started when the server has not created an active attempt.

ACCEPTANCE / PRECAUTIONS

- A student can explain when an answer is saved and when an attempt becomes final.
- The rehearsal does not contaminate analytics, rankings, or actual attempt responses.

The desktop examination workspace

FOCUS ON THE QUESTION

Implement the exam shell as a dedicated workspace with stable controls and no unnecessary competing content.

Layout and reading

Use a compact top bar for paper identity, the approved timer display, connection and save status, and a Help or instructions control. Place the current section and question identifier above the main question canvas. The desktop palette occupies the right panel when width allows. Keep the answer controls and profile-permitted navigation actions near the question, with enough space that long diagrams do not overlap them.

Render the full question stem, options, tables, formulas, units, and figure captions. Preserve meaningful reading order. Allow students to zoom an image in an accessible viewer without changing answer state. The question number in the interface is a display position; the backend uses a stable question identifier. Do not load or reveal the answer key in the exam client.

Answer interactions

For single-choice questions, support exactly the selection behavior of the configured type. Multi-select and numeric controls appear only when that profile and question schema permit them. Use visible labels, focus states, and error messaging. Input validation distinguishes an invalid format from an incorrect answer; correctness is withheld until the mode allows release.

Use Save and next, Clear response, Mark for review, and navigation controls only in the combinations defined by the profile. An answer and a review marker may coexist if that is permitted. Never represent review status as a substitute for a response value. The transition engine in Volume 2 decides the authoritative state.

Distraction and stability rules

No advertisements, upsells, achievement popups, public chat, recommendation panels, or celebratory animation appear during a simulation. Do not shift the content vertically when a save succeeds. Keep the question scroll position stable while updating background indicators. Refresh recovers the permitted authoritative attempt state rather than reconstructing it from a visual status color.

Copy-ready implementation direction

Build the shell with fixtures for a short MCQ, long derivation, scanned figure, numeric response, missing asset, and section-specific restriction. Demonstrate the same server-side attempt through answer selection, revisit, reload, and submission. Verify the recorded response exactly matches the selected option or submitted value and that the interface does not imply a successful save before acknowledgment.

ACCEPTANCE / PRECAUTIONS

- Long mathematics and figures remain readable without overlapping timer or actions.
- The application never exposes unreleased correct answers in exam HTML, bundles, or API payloads.

Question navigation and status semantics

THE PALETTE IS A VIEW OF STATE

Students must be able to locate unfinished work without interpreting hidden status rules or losing a response during navigation.

Palette model

Use the approved profile to determine which statuses exist and how they map to saved responses and review markers. A common product display may distinguish unseen, seen without response, answered, marked for review, and answered plus marked for review when those states are permitted. The authoritative mappings, transitions, and scoring consequences come from Volume 2; do not assume this display vocabulary applies unchanged to every exam.

Every palette cell has a visible question number, an accessible label containing position and status, a focus state, and a distinct current-question indicator. Include a persistent legend or an easily reachable legend control. For many questions, use a scrollable palette that keeps the current section clear and does not trap keyboard users.

Moving between questions

Changing questions follows the configured navigation policy in Volume 2. Distinguish unsaved drafts from responses committed by an explicit SAVE or REVIEW action permitted by the profile. If navigation discards a draft under that policy, explain the consequence in instructions and the unsaved indicator; preserve the committed answer. Background draft recovery is never an exam commitment. Handle pending commits under Volume 5.

Where section access is restricted, unavailable destinations explain the reason. A profile may allow returning to a question, forbid certain backtracking, or impose section timing. Implement those behaviors from the configuration rather than disabling controls cosmetically. Student-visible instructions and server enforcement must agree.

Keyboard and visibility

Provide predictable keyboard access to the palette, answer controls, and navigation actions. Announce question changes without repeatedly reading the entire dashboard. Do not steal focus when a background save completes. On narrow screens the palette opens as a drawer with an accessible close action and returns focus to its trigger.

Show section-level counts only from current authoritative state. If an event is still awaiting acknowledgment, separate its pending indicator from the confirmed count. Do not round or estimate the number of answered questions.

Verification direction

Copy this test requirement: Exercise every allowed transition for each profile, including selecting then clearing, answering then marking review, viewing without answering, revisiting a saved numeric answer, and attempting a forbidden transition. Compare the resulting palette, saved response record, and scored result against the same state-transition specification.

ACCEPTANCE / PRECAUTIONS

- No palette state is meaningful only through color.
- Rapid navigation does not lose an answer or create a false confirmed count.

Timing, save feedback, and recoverable interruption

TRUST MUST BE VISIBLE

Make the timer and save system trustworthy through clear status and recovery behavior. A visible countdown alone is not an examination clock.

Timer presentation

Display time remaining using the authoritative attempt policy and server time reference described in Volumes 2 and 5. Keep its location stable. Show low-time warnings only at configured thresholds, with text and accessible announcements that do not continually interrupt screen readers. A browser clock change, hidden tab, or refresh must not reset or extend the allowed time.

If a mode supports pause, make the paused state and restart conditions explicit. If it does not, a connection interruption must not appear to pause the examination. Do not offer controls that conflict with the approved rules. Show any authorized accommodation in the instructions and stored attempt metadata.

Save-state vocabulary

Use Unsaved selection for an uncommitted draft. Use Saving response only while an explicit SAVE or REVIEW commit request is pending, Response saved after server acknowledgment, Connection lost - response pending when appropriate, and Save failed - retry available when recovery requires action. Distinguish recoverable drafts, queued commits, and server-confirmed responses. Avoid a green saved badge driven only by a local browser write.

Keep pending counts small and relevant. When several commits are queued, state how many await acknowledgment. A failed numeric validation identifies the syntax problem without implying that the answer is academically wrong. Do not permit UI retries to reorder or overwrite newer responses with stale requests.

Interruption and return

After a refresh, device reconnection, or session renewal, fetch the authoritative attempt and reconcile permitted pending commits according to the conflict policy. A recovered draft remains uncommitted. Show Restoring attempt state until this finishes. If an attempt expired or was already submitted, present its terminal state and explain where queued changes stand.

If using multiple devices is restricted, show a clear message and follow a tested ownership policy. A new device must not silently create a second simultaneous writable copy. Recovery should not require a student to guess which answers were saved.

Submission feedback

The final confirmation summarizes the consequences and relevant response counts. After confirmation show Submitting attempt until the server confirms completion. Then display receipt time and result availability. A network failure does not produce a fake success page; safe retry uses the same submission identity.

ACCEPTANCE / PRECAUTIONS

- Reload, offline recovery, and duplicate submission demonstrations show consistent server records.
- Students can tell pending, confirmed, expired, and submitted states apart without developer tools.

Mobile and narrow-screen examination experience

RESPONSIVE WITHOUT CHANGING THE RULES

Support phones through explicit design and verification. A smaller screen must not introduce different scoring, timing, or response persistence.

Narrow-screen layout

Keep the timer, current question identity, and save state visible in a compact top bar. Stack the question content and answer controls. Move the palette into an accessible drawer or sheet with a visible trigger and current-status summary. Use a sticky action row only when it does not cover content or the browser keyboard. Allow scrolling through long questions without trapping the student between fixed bars.

Use sufficiently large touch targets and spacing so Clear response and Save and next cannot be confused by a small movement. Meet the applicable WCAG 2.2 target-size requirements and test actual controls [V1-S1]. Do not rely on swipe gestures for essential actions. A swipe may be an optional shortcut only if it is safe, discoverable, and consistent with the selected profile.

Mathematics and figures

Responsive equations must remain legible and keep semantic order. For an equation or table that genuinely needs more width, offer controlled horizontal scrolling inside that content region and a visual cue. Do not force the entire page to scroll sideways. Figures may open in a zoomable viewer with an accessible caption and close action.

Preserve aspect ratio and prevent important labels from being cropped. If a scan is too blurred for a phone, flag a content-quality issue rather than claiming responsiveness fixes it. For numeric answers, select a useful keyboard without relying on it to enforce valid syntax.

Device interruptions

Test rotation, browser address-bar changes, software keyboard appearance, incoming backgrounding, connection changes, and memory-related reloads. Returning to the page applies the same recovery and timer policy as desktop. A mobile browser tab must not receive unlimited time because animation frames paused.

Avoid opening new tabs for core examination controls. A solution viewer after submission may use a dedicated route, but the student should be able to return to the same review position.

Verification direction

Copy this requirement: Test real narrow-screen behavior using representative Android and iOS browsers, an ordinary laptop, and a tablet. Confirm the full path from onboarding to final review with actual data. Include long mathematics, tall diagrams, large text, and failed saves. Emergent recommends real end-to-end and device checks before sharing [V1-S2].

ACCEPTANCE / PRECAUTIONS

- At narrow widths, the software keyboard never hides the active numeric input or its essential action.
- The same saved attempt produces the same result across supported devices.

Results overview and the first useful explanation

LEAD WITH MEASURED OUTCOMES

The first result screen should explain the attempt clearly before asking the student to explore detailed charts.

Overview composition

Show paper identity, mode, profile version, attempt date, submission state, and whether the result is final or provisional. Place the measured score and its maximum or applicable scale first. Follow with subject or section contribution, correct and incorrect counts, unattempted count, time use, and one concise next-step panel. Every value must use the definition in Volume 4 and the scoring rules in Volume 2.

Use a small number of meaningful cards instead of a wall of 60 metrics. A card should show value, unit, denominator, definition link, and data status where relevant. Keep charts consistent with table values. Explain negative marks and unanswered questions using the configured rules rather than a generic interpretation.

Result authority and update states

Show Pending evaluation when the server has accepted the attempt but scoring is not yet available. A failed analytics job does not invalidate a correctly stored submission. Allow the score to appear separately from later analyses if the workflow supports that separation. Clearly distinguish unavailable analysis from zero.

If an approved answer-key correction creates a regrade, show the report revision and a concise explanation of changed outcomes. Preserve access to the change history where permitted. Do not overwrite a previous score without recording the correction and notifying affected students through the approved product channels.

Percentile and comparison placement

Display platform percentile only when the eligibility and evidence rules in Volume 4 are satisfied. Put the cohort label, eligible sample size, paper or comparability scope, and computation time near it. Show an explanatory unavailable state for a small, incompatible, or contaminated cohort. Never relabel this number Official NTA percentile or national rank.

Keep comparison optional and secondary to learning. If a leaderboard is included, its visibility and identity rules must be explicit and student-controlled. A result report remains useful without a public ranking.

Calls to action

Offer Review questions as the main next action, then Explore analyses and Save next steps. Provide a clean printable or exportable report if included in the approved scope. Exports repeat mode, version, cohort labels, and unavailable states so a downloaded file cannot misrepresent the result.

ACCEPTANCE / PRECAUTIONS

- Scores and counts reconcile with the stored response-level scoring ledger.
- The report never replaces missing evidence with a reassuring invented metric.

Question-wise review and solution presentation

CONNECT EACH ERROR TO EVIDENCE

A student should be able to inspect any question, understand the recorded response, and follow its approved solution without losing review context.

Question review layout

Show original stem and figures, recorded final response, approved answer, outcome, awarded marks, response or review status, and relevant time measurements with their interpretation limits. Include subject, chapter, concept, question type, difficulty source, and content revision tags. Use a clear visual distinction between Your answer and Approved answer; correctness is also written in text.

Allow filters for correct, incorrect, unanswered, marked review, subject, chapter, and supported issue categories. Navigate previous and next within the current filtered set. Keep the student's place when returning from a solution image or related question. Stable question identifiers prevent review from pointing at the wrong item after ordering changes.

Solutions from supplied PDFs

Attach an approved question-to-solution mapping, not merely the entire solution PDF beneath every question. Present the explanation as readable structured content when reliable extraction is approved, and retain a source-page crop or PDF reference for provenance. Preserve formulas, units, diagrams, and the logical order of steps. When only a reviewed page image is available, label its format accurately.

The answer key and the worked solution may disagree. Volume 3 defines the editorial resolution workflow. Students must not see an unresolved contradiction presented as authoritative. If a solution is unavailable, say Solution not yet approved rather than generating an unchecked explanation on demand.

Student annotations and issue reports

Permit private bookmarks and notes if included in scope. A structured issue report can identify unclear question, suspected wrong answer, mismatched solution, poor figure, or missing explanation. Include the question and attempt identifiers automatically. A report creates a review item; it does not change the student score immediately.

Learning suggestions may point to an approved concept explanation or another practice set. Do not tell a student that a single wrong answer proves a stable weakness. Separate observed errors from suggested causes.

Question-level cohort information

Show empirical solve rate, option distribution, or time comparison only when the Volume 4 evidence gates pass. Question-wise percentile refers to a specifically defined comparable performance metric, not a percentile inherent in the question. Explain exclusions and missing evidence near the chart.

ACCEPTANCE / PRECAUTIONS

- Each solution belongs to the correct stable question identifier and reviewed content version.
- A submitted issue can be tracked without exposing private notes to other students.

Analytics navigation and interpretation

60 DEFINITIONS, A SMALL NUMBER OF CLEAR VIEWS

Use the Volume 4 analytics registry to organize analysis. Do not invent additional scores merely to fill the result dashboard.

Information architecture

Group analytics into Score and coverage, Accuracy and response choices, Time and pacing, Topic and skill evidence, Strategy and review behavior, Cohort comparisons, Progress over attempts, and Recommendations. These groups are navigation containers, not separate metrics. The registry defines which individual analyses count toward the required minimum of 40 and the complete target of 60.

Open the most relevant group from the result overview or question review. Provide a persistent paper and attempt context, then allow permitted filters by subject, chapter, question type, and time window. A longitudinal view shows exactly which attempts are included and why. Do not combine custom learning and full simulation attempts by default.

Metric card contract

Each analytic panel exposes a plain-language definition, calculation basis, denominator or population, unit, eligibility, exclusions, version, and unavailable reason where needed. Include an expandable calculation example for advanced metrics. Charts have accessible summaries and a table alternative. A tooltip alone is insufficient for crucial caveats.

Values must come from the tested analytics layer. The interface never computes a competing percentile formula from a convenient subset of browser data. Repeated visualizations, renamed totals, and sorting the same score by a different color do not count as new analyses.

Risk, precaution, and interpretation

Risk means a defined educational or strategy observation, such as repeated negative-mark exposure or poor time allocation, with evidence and a proportionate action. It is not a diagnosis of personality, intelligence, anxiety, or future success. Precautions are specific: practice response validation, reserve review time, or verify a known unit conversion, when the recorded data supports the suggestion.

Mark uncertainty where evidence is sparse or time telemetry is incomplete. One session does not establish a reliable trend. A graph across incomparable papers must explain normalization or show the series separately. Avoid false precision in recommendation confidence.

Copy-ready requirement

For each implemented analysis, show a demonstration record with the underlying inputs and the expected output from a reproducible fixture. Show zero, insufficient-data, filtered-empty, and permission-restricted cases. Let administrators see analytics job health without exposing extra student data. Complete definitions and tests before decorative chart styling.

ACCEPTANCE / PRECAUTIONS

- At least 40 distinct registry analyses work, with honest unavailable states where evidence is absent.
- A student can identify the cohort and denominator behind any displayed comparative number.

Question library and an actionable study planner

TURN REVIEW INTO A SMALL NEXT STEP

The platform should help students act on results without creating a complex productivity application they must maintain.

Private question library

Collect bookmarked questions, previously incorrect questions, unanswered reviewed questions, and questions assigned through approved practice sets. Let students filter by exam family, subject, chapter, concept, question type, and last outcome. A saved question references its stable content identifier and approved revision. If content is corrected or withdrawn, reflect that status and retain the explanation needed to understand prior attempt records.

Separate private notes from platform explanations and public content. A student can remove a bookmark without deleting attempt history. Search shows only questions the student is entitled to view. Questions from unreleased papers or active attempts with hidden solutions must not become visible through the library.

Planner basics

Offer a short list of concrete actions based on recent review: revisit a small set of mistakes, complete one concept practice set, or take the next comparable simulation. Each action includes the evidence that generated it, an estimated commitment if justified, and an easy completion or dismiss control. Use optional dates and reminders only if implemented and permitted.

Support student-created tasks such as revise integration methods or attempt WBJEE practice. Keep planner items separate from inferred recommendations. Recommendations should not silently overwrite a student's plan. Allow snooze, dismiss with a reason, and mark complete without claiming mastery from a checkbox.

Personalization limits

Avoid unsupported promises such as Your score will improve by 20 marks if you complete this task. Explain suggestions with observed evidence and conservative language. If a chapter has too few questions, say More evidence is needed rather than classifying it as mastered or weak. Retakes may reflect memory; longitudinal comparisons must treat that explicitly.

A recommendation engine can begin with transparent tested rules. Do not add opaque AI ranking merely to claim personalization. Advanced model-generated plans require the validation and audit standards in Volume 4.

Implementation priority

Copy this direction: Launch bookmarks and a simple next-action list after the exam and review flows work. Keep a richer planner, reminders, mentor collaboration, and spaced-repetition scheduling as later increments. Show an actual mistake flowing from submitted result to library to a new approved practice session, with permissions preserved across the entire flow.

ACCEPTANCE / PRECAUTIONS

- Private notes never appear in another student account or public cohort analysis.
- A suggested task states its evidence and remains editable or dismissible by the student.

Administrator screens and editorial control

MAKE PDF REVIEW PRACTICAL

Administrators need an efficient workbench that surfaces uncertainty. Publication quality cannot depend on remembering a hidden checklist.

Import workbench

Provide an upload screen for question PDF, answer-key PDF, and solution PDF, allowing the answer key and solutions to share one source when appropriate. Ask for examination family, candidate profile, language, paper title, and source rights declaration or record as required by the deployment policy. Show upload status, processing stages, and meaningful extraction errors.

The review screen pairs the source page or crop with extracted question content. Highlight low-confidence boundaries, option parsing, formulas, diagrams, missing answers, ambiguous mappings, and duplicate candidates. Editors can correct text, preserve page references, split merged questions, and link the correct solution. Volume 3 specifies the data and pipeline behavior.

Review and publication

Show the paper inventory with draft, processing, needs review, approved, published, corrected, and withdrawn statuses as applicable. Publication requires the explicit checks in Volumes 2 and 3: profile compatibility, completeness, answer-key validation, solution mapping where provided, figure readability, permitted composition, and reviewer sign-off. Display blockers and warnings separately.

Require a preview in the student exam shell before publication. Administrators can take a non-public verification attempt and inspect the scoring ledger. The preview is excluded from student analytics and comparisons. Editing a published paper creates a revision workflow instead of silently mutating an active attempt.

Profile and operational screens

The profile editor shows official source links, effective year, rule fields, reviewer decision, and the impact of a proposed change. Keep unverified rules visibly unpublished. Cohort tools show inclusion policy, eligible sample size, contamination checks, and metric availability; they do not manufacture a high percentile for marketing.

Support screens search permitted account and attempt identifiers, show save or submission status relevant to the incident, and record support actions. Mask unrelated personal information. Audit screens show who changed, approved, published, withdrew, or regraded records.

Copy-ready requirement

Build administrator interfaces with actual review states, permission checks, and an audit trail before importing a large corpus. Demonstrate one imperfect PDF progressing through flagged extraction, human correction, question-to-solution mapping, approval, preview, and publication. Include a failed publication attempt so reviewers can see that blockers are enforced server-side.

ACCEPTANCE / PRECAUTIONS

- Unapproved imports never reach the student catalog or answer-key endpoints.
- An editor can identify and resolve every blocking issue from the workbench without editing the database manually.

Accessibility, language, and permitted accommodations

INCLUSION THROUGH TESTED BEHAVIOR

Target WCAG 2.2 Level AA for the supported application scope. Treat conformance as verified behavior, not a badge added after styling.

Core accessibility contract

Use semantic headings, labeled controls, meaningful link text, table headers, accessible dialog behavior, and logical focus order. Provide visible focus that is not hidden by sticky elements. Verify contrast for text, controls, focus, and meaningful graphics. Support keyboard operation without requiring a pointer. Avoid using color, location, shape, or sound alone to communicate an essential instruction [V1-S1].

Check text resizing and reflow. Keep essential reading and actions usable at zoom. Apply appropriate target-size requirements and spacing. Respect reduced-motion preferences and avoid flashing, autoplay, or nonessential movement. Test with automated tools and manual keyboard and screen-reader workflows; an automated score does not prove full conformance.

Mathematics and figure access

Use accessible mathematical rendering where practical, with semantic source and tested assistive-technology behavior. Supply reviewed text descriptions for essential figures where those descriptions can preserve the question's meaning without supplying the answer. Do not invent a diagram description from an unreviewed model output. Content accessibility must be part of the editorial record.

If a figure-dependent question cannot be made accessible in a mode, document that limitation, identify appropriate reviewed alternatives, and handle accommodation through the approved policy. Do not claim that adding generic alt text makes every mathematical diagram equivalent.

Localization

Separate interface strings from question content. English interface is the initial default; other interface languages use reviewed translations and preserve numerical, mathematical, and proper-name meaning. Question language options correspond to approved content versions. Never automatically translate a live exam question and present it as official or editorially approved.

Keep units, numeric syntax, decimal separators, answer validation, and locale formatting compatible with the exam profile. A language preference must not change scoring rules. Long translated labels must fit buttons, palette descriptions, notices, and mobile layouts.

Accommodations

Support approved accommodation settings such as additional time only through the explicit profile and authorization workflow. Snapshot them before start, display their consequences, and retain them in comparison eligibility. Timing-sensitive exam content may have applicable accessibility exceptions; review and document them rather than silently altering an official simulation to meet a generic default.

ACCEPTANCE / PRECAUTIONS

- Complete an attempt and review using keyboard alone and a representative screen reader.
- Claims of accessibility identify tested scope, known limitations, and the current verification date.

Error language, empty states, and product trust

WRITE WHAT ACTUALLY HAPPENED

Use short, specific messages that explain the state and the next safe action. Avoid blaming the student for technical failures.

Exam and result messages

Use Connection lost. Your latest response is waiting to sync. The exam timer continues under this profile. Use Response saved only after server confirmation. Use We could not confirm submission yet. Retry submission; your attempt ID is unchanged when idempotent retry is available. Use This attempt has ended. Open the result or submission status instead of presenting an editable expired paper.

For analytics, use This comparison needs more eligible attempts, Time analysis is incomplete, or No questions match these filters. Explain missing values separately from zero. A missing solution says Solution not yet approved.

Catalog and import messages

Use No published papers match these filters with Clear filters. Use This paper was withdrawn; choose another available paper with a catalog link. Use This PDF is readable, but several question boundaries need review for a partial extraction. Use Publication blocked: three questions have no approved answer rather than a generic validation failure.

Associate errors with their affected action or record. Preserve entered data and source files when a recoverable operation fails. Do not ask a user to upload the same large PDF again if the server still has a valid uploaded source. Provide a support reference that contains no sensitive answer or personal data.

Trust-building copy

Call simulations Practice simulations and comparison values Platform percentile or the exact approved label. Explain that exam rules are based on the displayed official sources and profile version. Do not call the platform an official examination portal. Marketing claims about student counts, success rates, speed, or capacity require actual measured evidence and scope.

Recommendations use language such as Your recorded errors suggest reviewing this concept or In this attempt, later questions received less time. Avoid You are careless, Your intelligence is low, or You will fail. The interface should support improvement without making unsupported judgments.

Copy review procedure

Copy this requirement: Keep messages in a shared string catalog with states, triggers, and recommended actions. Test long translations and screen-reader announcements. During review, provoke each failure with controlled fixtures and verify that the message matches the server state. Remove placeholder toasts and generic Something went wrong where a useful specific explanation is available.

ACCEPTANCE / PRECAUTIONS

- Every important failure offers a safe next step and preserves recoverable work.
- No message describes locally queued work as confirmed or a practice statistic as official.

Implementation priorities and the product acceptance gate

PROVE THE COMPLETE PATH

Build in stages that produce a usable vertical slice. A polished landing page is not evidence that the examination platform works.

Stage priorities

Stage 1 establishes one approved profile, one reviewed paper, authentication, the responsive exam shell, answer persistence, timing, submission, scoring, and question review. Stage 2 adds the general PDF workbench, the remaining approved exam profiles, custom integer practice, publication controls, and basic results. Stage 3 completes at least 40 registry analyses, cohort evidence gates, progress history, bookmarks, and simple next actions. Stage 4 completes the 60-analysis specification and advanced features after evidence supports them.

Across all stages, treat authorization, answer-key confidentiality, accessibility, content correctness, and recoverable failure as foundational. Volume 5 expands implementation and capacity gates. Hundreds of thousands of registered students is a scale intention; release claims must describe the simultaneous load actually tested.

Demonstrable product acceptance rubric

Require ten demonstrations: create an account and choose an available paper; review its exact profile; complete instructions; answer every supported type; navigate every permitted status; recover from a reload and connection loss; submit once under duplicate requests; reconcile scoring with the approved key; inspect a correctly mapped solution; and follow a supported next action. Each demonstration includes the browser behavior and the relevant authoritative records.

Then require administrator evidence: upload imperfect PDFs, correct flagged content, resolve an answer/solution conflict, block invalid publication, approve and publish a revision, and verify access restrictions. Show phone and keyboard workflows. Provide measured end-to-end checks before sharing; use the Emergent review guidance as supporting workflow, not as a substitute for this acceptance contract [V1-S2].

Rubric for readiness

The release is ready only when required journeys pass, no critical or high-severity defects remain, the source/profile/content/analytics versions are traceable, data-backed values reconcile, and recovery does not lose confirmed responses. A partial feature has a visible unavailable state and does not produce fabricated data. Label unfinished development screens and keep them outside public navigation.

Final prompt to Emergent

Show the working release, evidence for every binding requirement, the latest decision log, the tested device matrix, and known limitations. Separate completed capabilities from planned ones. Ask for the next implementation stage only after the current slice is reviewable. Keep the interface simple and the correctness evidence detailed.

ACCEPTANCE / PRECAUTIONS

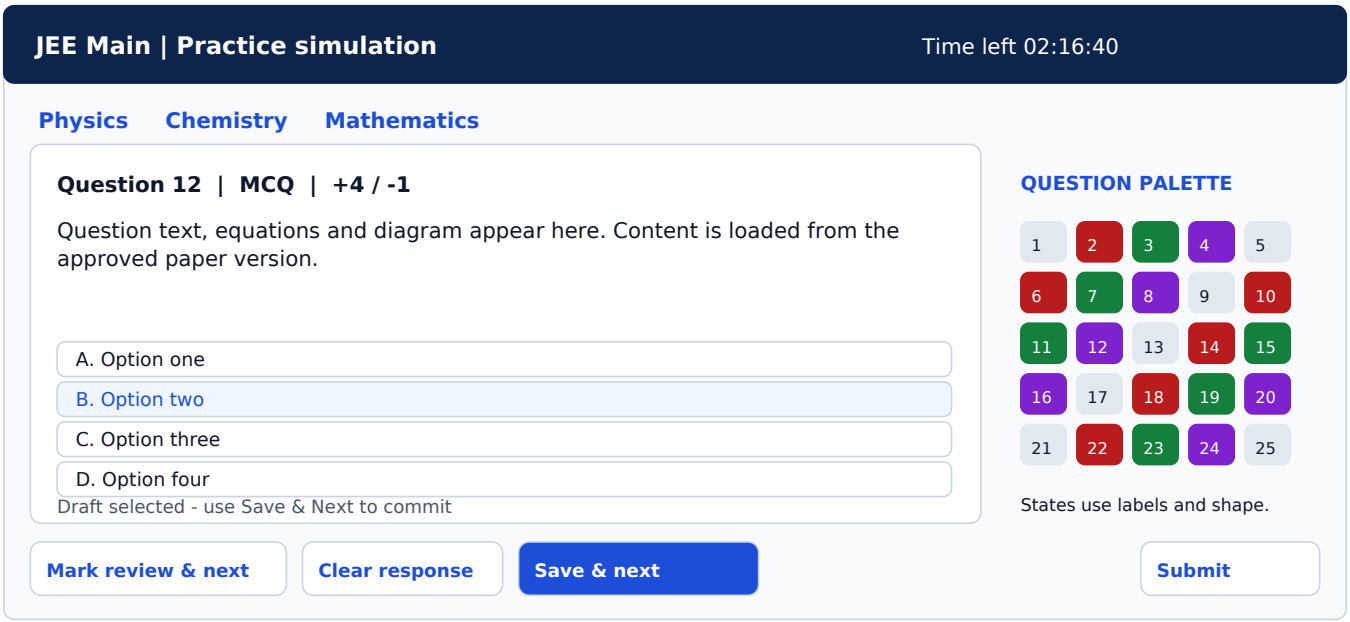
- Release review covers actual student and administrator journeys using persisted application data.
- The owner receives a concrete checklist of passed requirements and unresolved blockers, not a claim that everything is production-ready.

Desktop exam workspace: layout blueprint

VISUAL ACCEPTANCE REFERENCE | PROPOSED JEE MAIN PRACTICE SHELL

Use this layout to build a calm, readable desktop exam workspace. The diagram specifies hierarchy and control placement; Volume 2 determines the exact official-baseline behavior for each configuration.

ANNOTATED STRUCTURE REFERENCE - PROPOSED PRODUCT LAYOUT



Values and content shown are illustrative; approved configuration and backend data control the live interface.

Build the interaction behind every control

Keep the timer, current subject and save status visible. Show one question at a time with large mathematical content and a question palette beside it. Differentiate a local selection from a server-acknowledged committed response. Palette navigation must follow the configured draft policy; it must never silently convert an unsaved selection into a scored answer.

Adapt the shell to the selected exam

Load the palette legend, answer controls, section labels, marks and review actions from the immutable configuration. Replace radio buttons with checkboxes for WBJEE Category III and with the approved numerical keypad for JEE numerical items. A BITSAT bonus-entry confirmation is a separate irreversible dialog. Reuse visual primitives while keeping each rule explicit.

ACCEPTANCE / PRECAUTIONS

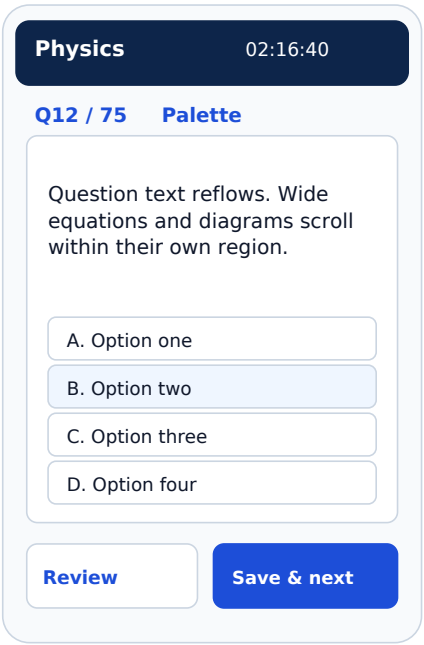
- At 1280-pixel width, the question and palette fit without page-level horizontal scrolling.
- At 200% zoom, controls remain reachable and focused content is visible.
- The selected answer, saved answer and review state have clear text equivalents.

Phone exam workspace: layout blueprint

VISUAL ACCEPTANCE REFERENCE | RESPONSIVE PRACTICE MODE

Implement mobile practice as a deliberate responsive layout. Keep the familiar rules but acknowledge that a phone cannot reproduce an examination-center workstation. Give students a usable experience without obscuring mathematical content.

ANNOTATED STRUCTURE REFERENCE - PROPOSED PRODUCT LAYOUT



1. Keep status visible

Timer and save acknowledgement stay visible without obscuring the current question.

2. Use a drawer for navigation

Palette opens on request and closes after a deliberate jump. Restore focus to the chosen question.

3. Protect touch actions

Primary controls have generous touch targets. Confirm final submission and any irreversible transition.

4. Preserve question content

No full-page horizontal scrolling. Zoomable figures retain labels and accessible descriptions.

Values and content shown are illustrative; approved configuration and backend data control the live interface.

Navigation and input behavior

Use an on-demand question-palette drawer and a compact subject selector. When a drawer closes, return focus to the chosen question. Position the primary save action consistently. Let the operating system keyboard open without covering numerical entry or the current save state. Suppress accidental form submission by Enter unless the active exam configuration explicitly maps that key.

Content and device checks

Reflow normal prose at a narrow viewport. Give long equations and figures their own pan or zoom region, with accessible text and reset controls. Re-check the layout on Android Chrome, iOS Safari and a foldable screen. Rotate, background and resume the app while the server deadline continues. Never depend on mobile background timers for finalization.

ACCEPTANCE / PRECAUTIONS

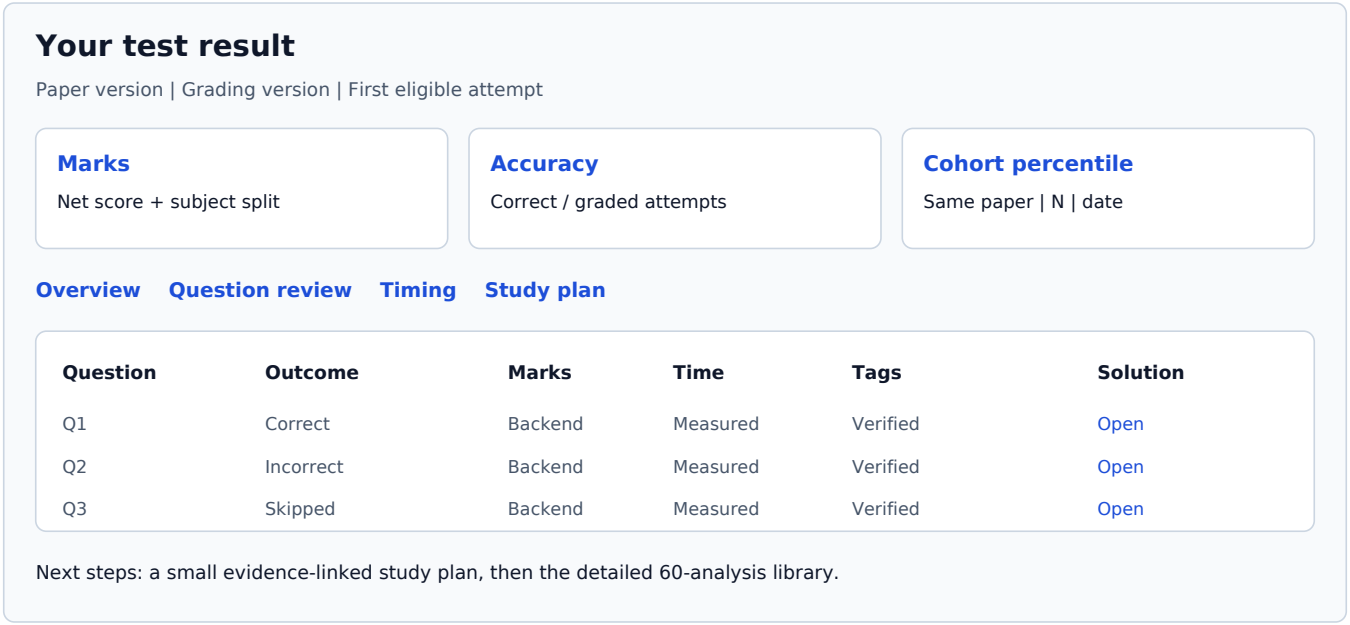
- At a 360-pixel viewport, essential controls and the timer remain visible.
- Opening the keyboard does not hide the current numerical input or commit control.
- A reconnect banner explains unsent changes and the authoritative deadline.

Result dashboard: layout blueprint

VISUAL ACCEPTANCE REFERENCE | PROGRESSIVE DISCLOSURE OF 60 ANALYSES

Make the first result screen easy to understand. Present the score, accuracy and an eligible cohort comparison before the full analysis library. Every displayed number must come from a documented, versioned backend computation.

ANNOTATED STRUCTURE REFERENCE - PROPOSED PRODUCT LAYOUT



Values and content shown are illustrative; approved configuration and backend data control the live interface.

A readable first screen

Start with marks and subject splits, then show attempted, correct, incorrect and skipped counts. Put a short explanation beside the platform cohort percentile, including eligible N and the snapshot date. If no valid cohort exists, show an unavailable state instead of a fabricated comparison. Use a compact tab layout for overview, question review, timing and study plan.

Question-wise evidence and next actions

The question review table must link the committed response, award, measured timing, verified tags and released solution. Selecting a row opens the full question and explanation, preserving the result filters. Include a small, evidence-linked plan, then make all 60 analyses searchable by category. Keep formulas, coverage limits and revision lineage available on demand.

ACCEPTANCE / PRECAUTIONS

- Cards, question rows and downloaded reports reconcile with the same grading version.
- No official rank, admission guarantee or NTA-normalized percentile is inferred from a mock cohort.
- A student can move from a weak topic to the relevant question evidence and revision action.

Shared contract across all five volumes

CANONICAL NAMES | API PREFIX | RULE PRECEDENCE

Treat the five volumes as one specification. Resolve wording differences through the following shared contract before implementing APIs, migrations or scoring. Record any deliberate deviation in the project decision log.

Canonical entities and naming aliases

Use Question (also called Item) for the reusable bank identity and QuestionVersion (ItemVersion) for an immutable content snapshot. ExamConfig or ExamProfile describes the versioned rules; an Adapter is the code that interprets those rules. PaperVersion is the published ordered manifest of question versions, configuration version, solution versions and release policy. Attempt binds one student to that immutable manifest. GradingVersion records the scoring implementation, key revision and correction lineage. Internal naming may differ, but every migration and API must map these concepts consistently.

One API contract and two response states

Use /v1 as the canonical API prefix and publish one OpenAPI specification. Short /admin, /attempts or /uploads routes in worked examples are endpoint families under /v1, rather than separate unversioned APIs. Separate draft_response from committed_response. Background persistence may recover a draft, but only a profile-eligible explicit commitment event changes the response eligible for scoring. Show local edit, durable draft save and committed-answer acknowledgement as distinct states. Finalization freezes the eligible committed responses; it must not invent a missing Save or Review action.

Precedence and change control

A current official rule source establishes the relevant exam baseline. The administrator-approved immutable exam configuration implements that source. Volume 2 defines its executable interpretation; Volume 3 binds reviewed content to it; Volume 4 derives analysis from accepted attempts; Volume 5 preserves those contracts in production. Volume 1 controls product language and presentation. Custom practice overrides must be labeled and versioned. If a verified source conflicts with a proposal, correct the proposal, publish a new approved version and demonstrate regression fixtures. Do not change rules for attempts already running.

ACCEPTANCE / PRECAUTIONS

- Generate typed clients from the canonical OpenAPI contract and validate all example endpoint families against it.
- Prove that a durably autosaved draft remains unscored until the required commitment action.
- Before launch, publish a traceability matrix linking requirement, configuration, implementation and test evidence.

Source register and implementation handoff

EVIDENCE AND SCOPE OF THIS SPECIFICATION

Use these references to verify the stated baseline. Requirements and engineering thresholds in this volume are proposed product decisions unless a source explicitly establishes them.

V1-S1 | W3C: Web Content Accessibility Guidelines (WCAG) 2.2

https://www.w3.org/TR/WCAG22/

Checked: 2026-10-12. Primary normative reference for the accessibility target. Apply specific success criteria and test actual application states. Source verified by the coordinating agent.

V1-S2 | Emergent: Try it before you share it

https://emergent.sh/docs/try-it-before-you-share-it

Checked: 2026-10-12. Primary product documentation supporting real end-to-end and device checks before sharing. Source verified by the coordinating agent.

V1-S3 | Emergent: Write prompts that work

https://emergent.sh/docs/write-prompts-that-work

Checked: 2026-10-12. Primary product documentation supporting clear purpose, core features, design feel, examples, a changelog, and incremental refinements. Source verified by the coordinating agent.

ACCEPTANCE / PRECAUTIONS

- Before release, retain the approved configuration, source document date, verification owner and test evidence. Re-check any rule that changes for a new exam year.